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Interm/Mentor

Annotated Source List

AI in Therapy. (2025, July 23). Best AI note writing assistants for physical therapists in 2025. AI in Therapy. https://aiintherapy.com/2025/07/23/best-ai-note-writing-assistants-for-physical-therapists-in-2025/

Summary

This blog article highlights 5 applications of AI-powered tools for recording notes and information for physical therapists. The post leans into the burden of managing the administrative aspect of physical therapy and provides 5 solutions to streamline note-taking, medical records, and creating SOAP reports, which is a healthcare practice of formatting a report by the subjective, objective, assessment, and plan. Scribenote can turn your conversations into specific formats like SOAP reports. Noterro is an all-in-one tool that takes care of EMR, which is a term for electronic medical records. Abridge records patient conversation to allow therapists to review notes after the patient leaves the clinic. These tools are advertised to save a huge amount of time for therapists while not encroaching on any human to human interaction.

Application to Research

This blog post demonstrates some AI practices that enhance the way therapists work by utilizing AI tools as assistants, by taking notes and compiling records. The blog even suggests that by using AI as an extra pair of eyes on records, it would make information increasingly accurate and error-free. In my research I would like to interview or gather data from therapists in my area to see how accurately and effectively as well as how often they use AI for note-taking.

AI documentation in physical therapy today. (2024). YouTube. https://www.youtube.com/watch?v=-qmUVteZYM8.

Summary

This video features a licensed physical therapist explaining how AI documentation tools are used in clinical settings. It highlights benefits such as time savings and improved organization, while also acknowledging the need for therapist review to prevent errors. Overall, the video emphasizes that AI assists rather than replaces therapists.

Application to Research

This source is helpful for my research because it provides a firsthand professional testimony rather than an academic perspective. It supports the claim that AI is most effective when used as an assistant under human supervision. Including this source shows evidence of real clinical adoption while also reinforcing the importance of therapist oversight and the need for human decisionmaking.

Aliferis, C., & Simon, G. J. (Eds.). (2024). Artificial intelligence and machine learning in health care and medical sciences: Best practices and pitfalls (Open Access ed.). Springer. https://opentechbook.com/book/artificial-intelligence-and-machine-learni ng-inhealth-care-and-medical-sciences/

Summary

This textbook explains the core principles of artificial intelligence, machine learning, and reasoning systems/decisionmaking algorithms. It provides the framework needed to understand how AI systems function and where their limitations arise. It explains AI advancements in modern healthcare.

Application to Research

This book is useful because it gives me a technical foundation to accurately explain how AI tools used in physical therapy operate. It helps prevent oversimplification and allows me to evaluate claims about AI replacing therapists. This strengthens the credibility of my argument since I will have a greater understanding of the topic as a whole.

Alshami, A., Nashwan, A., AlDardour, A., & Qusini, A. (2025). Artificial intelligence in rehabilitation: A narrative review on advancing patient care. Rehabilitación (Madr), 59(2), Article 100911. https://www.sciencedirect.com/science/article/pii/S004871202500 0313

Summary

This scholarly article reviews the use of AI and machine learning in rehabilitation and physical therapy. The authors discuss AI applications for personalized patient care, outcome prediction, and real-time monitoring of exercises. The author provides specific examples of how AI could be used as a tool, citing sensors, virtual reality, and robotic devices to analyze movement. AI is being developed in evaluating motor function, functional status, movement, and recovery.

Application to Research

This article supports my research by diving into the methods and technology required for AI to assist in rehabilitation. I will hopefully be dealing with personalized treatment plans, predictive analysis, remote monitoring, and rehabilitation robotics in my internship, and all of those support systems may be improved by AI and machine learning.

Alsobhi, M., Alsubaie, H., Alshahrani, M., Alzhrani, A., & Altwijri, A. (2022, October 20). Artificial intelligence application versus physical therapist for squat evaluation: A randomized controlled trial. Journal of Medical Internet Research. https://pubmed.ncbi.nlm.nih.gov/34518568/

Summary

This medical abstract summarizes the findings of a controlled trial that studied the effectiveness of AI models in teaching physical therapy exercises to patients compared to therapists. AI technology was satisfactory in detecting correct form and was comparable to a trained therapist. It struggled, however, in fixing incorrect form, concluding that AI has potential to improve practice patterns and patient outcomes with professional supervision.

Application to Research

This study improves both my personal concerns and benefits my research about the use of AI in the field of physical therapy. It provides concrete and quantifiable data that AI is useful in screening and evaluation, however it will not fully replace therapist supervision and communication. One question I have is if AI could be developed to coach patients virtually in the comfort of their homes.

Artificial intelligence in physical therapy education: Evaluating clinical reasoning performance in musculoskeletal care using ChatGPT. Musculoskeletal Care, 23(3), e70177. (2025). https://doi.org/10.1002/msc.70177

Summary

This study evaluates the performance of ChatGPT in physical therapy education by testing its ability to complete musculoskeletal clinical reasoning tasks. The authors compare AI-generated responses to expert-developed answers and clinical guidelines. Results show that ChatGPT performs well in structured reasoning tasks such as identifying diagnoses and treatment frameworks but struggles with nuanced decisionmaking judgment. The article emphasizes that AI functions best as a support tool rather than an independent clinician.

Application to Research

This source is particularly valuable because it directly addresses whether AI can replicate physical therapists' clinical reasoning, which is central to my research question. The study strengthens the credibility of claims about AI's limitations and strengths. I can use this source to argue that AI may support therapists in education and planning, but it cannot replace human judgment and especially in complex patient care situations.

Areias, A. C., et al. (2024). Predicting Pain Response to a Remote Musculoskeletal Care Platform Using Machine Learning. JMIR Medical Informatics, 12, e64806. https://medinform.jmir.org/2024/1/e64806/

<u>Summary</u>

This study evaluates an AI tool that is embedded into a musculoskeletal care program, indicating that AI models have already been adopted in many clinical settings. The study's findings exhibit the effectiveness of the AI model, as patients who were treated with the use of the AI system as well had a 30% chance increase in reduction of pain versus those who weren't. The model in the study utilizes machine learning and outputs predictive analysis that can tell the patient and therapist to what frequency they need to meet and if they can meet virtually if a physical session is not needed. The study concludes by clarifying that the tool is not meant to fully replace therapist judgment but instead allows for more informed decisions.

<u>Application to Research</u>

This study provides information to answer my research because it provides real-world applications of a tool that physical therapists are implementing in their practices right now. It is reassuring to hear that while the AI has the statistics to back its effectiveness, the model will not override therapist judgment. I believe that it is scary that an inaccuracy in the AI could sway a therapist to make a misinformed and detrimental mistake. In the future, I want to investigate more models like this and interview therapist users about how they believe these AI models are still learning and can be improved, as well as their outlook on how much therapists use them versus their own judgment.

DeSola, Dani. Personal interview.

<u>Summary</u>

This interview will discuss the impact of AI on current education. Dani Desola is a current student at Northwestern University pursuing her DPT degree. I will change my interview questions to focus on if her education includes AI and if her or her professors see future visions. I will continue this summary when the interview is completed.

<u>Application to Research</u>

This interview will open my eyes to the world of training and education for physical therapists. I want to see what kind of knowledge young future therapists are receiving and what practices they will bring with them into the career field. I plan to compare and contrast my findings with Dr. Siu and Dr. Tami Ford.

Effects of an AI-based physiotherapy educational approach on developing clinical reasoning skills: A randomized controlled trial. BMC Medical Education, 25, 1378. (2025). https://bmcmededuc.biomedcentral.com/articles/10.1186/s12909-025-1378-1

Summary

These study results examine the impact of AIassisted learning tools on physiotherapy students' clinical reasoning skills. Students using AIsupported case simulations demonstrated improved knowledge retention and confidence compared to those using traditional methods. The study also reports increased "self efficacy" when interacting with AI systems, though it cautions against overreliance on technology without instructor oversight.

Application to Research

This study helps support my research by showing that AI can enhance learning and reasoning in physical therapy education without fully replacing instructors. It reinforces the idea that AI may improve preparation and efficiency while still requiring professional supervision. I can use this source to discuss how AI might influence the future workforce by shaping how therapists are trained.

Ford, Tami. Personal interview.

Summary

This interview will discuss the limitations of AI in physical therapy. Dr. Tami Ford is my mentor and does not employ the use of AI much in her practices. She has a DPT (Doctor of Physical Therapy) degree and keeps updated, always looking for innovations in the field. I will investigate deeperinto why she is reluctant to use other tools and what would need to change for her to take advantage of AI. I will continue this summary when the interview is completed.

Application to Research

This interview is critical because it provides an opposite viewpoint to my other interviews and sources. Hopefully by hearing what advancements AI would need to make for it to make more sense in the clinic and combining it with the possibilities of AI's future in my interview with Dr. Siu, I can estimate if and when AI will start becoming implemented widespread among clinics that are still currently skeptical of AI.

Net Health. (2024, October 17). AI in physical therapy: The future of operations and patient care. Net Health. https://www.nethealth.com/blog/ai-in-physical-therapy-future-operations-patient-care

Summary

This article focuses on AI's impact on patient care and clinical operations in physical therapy. It explains how AI can help with tasks like evaluating patients, making treatment plans, setting goals, and tracking progress. The article also explores how AI technologies assist with operational tasks. This is important because it allows human therapists to focus on interacting with patients. The article also discusses ethical issues such as patient privacy, data security, and algorithmic bias. These concerns highlight the need for professional oversight.

<u>Application to Research</u>

This source strengthens my research by showing the ethical challenges of using AI in physical therapy instead of human therapists. It will help raise important questions about trust and privacy. Balancing the benefits and risks will be valuable as I look into how AI can support human therapists without fully replacing them in rehabilitation and patient care.

Kapica, P. (2024, August 29). Using pose estimation algorithms to build a simple gym training aid app. Medium. https://medium.com/@pawelkapica/using-pose-estimation-algorithms-to-build-a-simple-gym-training-aid-app-ef87b3d07f94

<u>Summary</u>

This blog article describes how pose estimation algorithms can build a simple AI-powered gym training app. The author explains how computer vision detects body positions and analyzes movements to give feedback on exercise form and fundamentals. The app acts as a digital training assistant by tracking repetitions and guiding users to correct form in real time through their mobile device. The article highlights the accessibility and potential for AI tools to complement traditional coaching without replacing human trainers.

<u>Application to Research</u>

This source supports my research by showing a practical example of how AI can assist in exercise evaluation. It demonstrates real-world applications of AI in monitoring movement and providing feedback, which parallels AI use in physical therapy to track exercises, improve technique, and prevent injury. I was very glad to stumble across this article because a therapist I had spoken to was unsure if AI was capable of these feats yet. I could use this to explore how AI-based movement tracking might supplement therapists' work in clinics or home programs, and how quickly they could be implemented.

Physical therapists' attitudes toward artificial intelligence diagnostics. BMC Medical Education, 25, 1751. (2025). https://bmcmededuc.biomedcentral.com/articles/10.1186/s12909-025-1751-6

<u>Summary</u>

This article investigates physical therapists' perceptions of AI diagnostic tools through survey-based research. The findings reveal cautious optimism, with therapists recognizing AI's potential for efficiency and accuracy while expressing concerns about data privacy, liability, and loss of professional autonomy. Many respondents emphasized the importance of maintaining therapist control over final clinical decisions.

<u>Application to Research</u>

This source is useful because it reflects how therapists themselves feel about AI integration, not just how researchers or developers describe it. It allows me to include professional perspectives in my research, showing that while AI adoption is growing, human therapists still want to remain central to care decisions. This supports my focus on AI as a supplement rather than a replacement.

Physical therapists' perceptions and attitudes toward artificial intelligence in healthcare and rehabilitation. Journal of Medical Internet Research. (2025). https://pubmed.ncbi.nlm.nih.gov/38644241/

<u>Summary</u>

This study falls into the qualitative side rather than data based quantitative studies. It explores physical therapists' experiences and concerns regarding AI implementation in rehabilitation. Common themes include ethical responsibility, patient trust, workflow changes, and the fear of losing skills. Therapists noted that AI can improve efficiency but stressed that patient motivation, empathy, and the ability to adapt remain exclusively human skills.

<u>Application to Research</u>

This article strengthens my project by addressing the ethical and emotional aspects of AI in physical therapy, which are often overlooked in technical studies. It helps me argue that AI cannot replace the interpersonal elements of therapy, reinforcing the importance of human therapists even as technology advances.

PtEverywhere. (2024, October 7). AI in physical therapy: Everything you need to know. PtEverywhere. https://www.pteverywhere.com/media/ai-physical-therapy

<u>Summary</u>

This article discusses the emerging role of artificial intelligence in physical therapy by highlighting uses such as predicting injury risk, optimizing rehabilitation protocols, and enabling virtual or remote therapy support. I learned that AI has an ability to analyze data from wearables and past patient outcomes to personalize care. It can also simplify documentation and administrative work. Although the author proves how AI demonstrates potential in increasing efficiency and accessibility, the article stresses that AI cannot replace the human and emotional elements of physical therapy like empathy and motivation/communication.

Application to Research

This article provides information to answer my research question because it provides real-world ways AI is already being integrated into physical therapy practice. It connects directly to my question of how AI improves accessibility and patient rehabilitation. It details how wearables and virtual platforms are being used for exercise feedback. Additionally, the article acknowledges AI's limitations, which reinforces the importance of professional supervision and the human aspects of therapy that I focus on in my research.

Revolutionize Physical Therapy with Artificial Intelligence. (2024). YouTube. https://www.youtube.com/watch?v=qINED-lBAnM.

Summary

This video provides an overview of how AI is currently being used in physical therapy clinics, including documentation automation, data analysis, patient progress tracking, and marketing. It features examples of AI-assisted workflows and also discusses how therapists can use technology to reduce administrative burden while maintaining quality patient care. The video includes Dave Kittle, who is a PT business owner.

Application to Research

This video will serve as a practical example of AI in action, complementing my other sources. It helps illustrate real implementations and makes my research more accessible. I can use this source to demonstrate how AI tools are already affecting daily clinical operations because the video covers a diverse range of topics like posture analysis and marketing and branding.

Roggio, F. (2024). A comprehensive analysis of machine learning pose estimation models in human movement sciences. Science Progress, 107(4), 00368504241600082. https://www.sciencedirect.com/science/article/pii/S2405844024160082

Summary

This article reviews machine learning models used for pose estimation in human movement sciences. It evaluates the accuracy and potential of different algorithms for real-world applications, such as rehabilitation and sports training. The study also addresses the challenges of real-time motion tracking, including sensor limitations, algorithm bias, and the need for human supervision. The article emphasizes that while AI can enhance monitoring and assessment, human oversight remains necessary for ensuring safety and accuracy.

Application to Research

This source is useful because it offers scientific evidence for using AI models to track human movement. It supports my research by showing that AI can monitor exercises accurately, though it does not replace therapists. I can use this study to talk about how reliable AI systems are, what their limits might be, and the ethical issues that come up when adding AI to rehabilitation or training programs.

Silva, A., & Costa, P. (2023). Designing interactive social robots for post-stroke rehabilitation: A human-centered approach. arXiv. https://arxiv.org/abs/2305.07632

<u>Summary</u>

This study presents a socially assistive robot powered by AI that is designed to support patients rehabilitating from strokes. The AI was developed with oversight from both therapists and stroke survivors. The AI integrates a neural network with algorithms that monitor patient exercise and provide corrective feedback. The study includes an evaluation of the system, which was tested by 15 former stroke therapy patients doing exercises. The AI scored an average accuracy of 0.81, which is on par with expert consensus. The AI was found to be an effective coach, but its recognition of movement errors was inferior to a trained therapist.

<u>Application to Research</u>

This source is very relevant to my topic because it demonstrates the active role of AI in rehabilitation, not just the passive, behind the scenes work. It actively proves AI's ability to effectively gather and give feedback by monitoring exercise quality and making decisions about the patient's performance. One limitation of AI is supposedly its ability to convey things to patients, however in this study, quite the opposite is true as the AI system acheives an accuracy of 0.81, which is considered above average. The implications of this in theory would be the widespread integration of systems like these in clinics. The study also raises questions about how the AI performs for different demographics of people and their abilities.

Siu, Joseph. Personal interview.

<u>Summary</u>

This interview will discuss Dr. Siu's findings in his article and where he plans to take his research. Dr. Siu was a part of an article I used abot evaluating ChatGPT's clinical reasoning. His expertise as a part of the Department of Health and Rehabilitation Sciences in the University of Nebraska Medical Center provides him with credibility and valuable insight. I will continue this summary when the interview is completed.

<u>Application to Research</u>

This interview will be super helpful to my research because it will be my first time speaking to a professional other than my mentor about this topic. He also has much more experience in testing AI's limits than my mentor, so I hope to gain more accurate information and more realistic speculation.

University of St. Augustine for Health Sciences. (2021). Artificial intelligence in physical therapy: Cool applications & fascinating implications. https://www.usa.edu/blog/artificial-intelligence-in-physical-therapy-cool-applications-fascinating-implications/

Summary

This article discusses the several leading AI tools for physical therapists. It compares and contrasts the various features and advantages as well as the limitations of the AI models and which areas they excel at. The main kind of AI tool used in the physical therapy field is note-taking assistants that utilize electronic medical records and dictation to provide summary or finding. The SWORD Health program helps patients strengthen at home while updating information. Physitrack utilizes 3D anatomy imagery and can be downloaded as an app on a phone. Kaia Health has a motion coach that tracks patients movement when performing exercises through a phone camera. Overall, these assistants tend to make solo exercise more efficient, allowing the therapist to dedicate more time and attention to patient care.

Application to Research

This article provides information to answer my research because it provides real-world applications and tools that physical therapists are implementing in their practices right now. The actual utilization of these tools demonstrates how AI isn't just a theoretical tool but rather a common, daily practice. I aim to further research which models are used in my local community and even at my internship site. I also want to find statistics to see how often these AI tools make mistakes when dealing with medical records and evaluating patient performance.